$$p(x) \begin{cases} cx^2 & 1 \le x \le 4 \\ 0 & x < 1 \text{ or } x > 4 \end{cases}$$

$$\int_1^4 cx^2 dx = 1$$

$$\frac{c}{3}(64-1) = 1$$

$$c\,\frac{63}{3}(21) = 1$$

$$c = \frac{1}{21}$$

$$\int_2^3 \frac{1}{21} \cdot x^2 dx$$

$$\Rightarrow \frac{1}{21(3)}(27-8)$$

$$\frac{19}{21(3)}$$

$$0.3015$$